



Mechanical Engineering (ME)

University of Michigan College of Engineering

ME 495 – Laboratory II

Lab 2: Flexible Torsional System

(Scenario)

Wolverine System Dynamics
Consultants for Automotive, Aero, and Space Technology
Advanced Industry Research Park
Ann Arbor, MI 48109



To: Powertrain Dynamics Group
From: Edgar Meyhofer and Pramod Reddy,
Directors, Wolverine System Dynamics
Date: Sept 4th, 2026
Subject: Drive Shaft Vibrations in the Gulo 3000

During the last regular planning meeting with Mesa Hybrid Vehicles Inc., a major vibration problem was identified in its new prototype hybrid electric model, the *Gulo 3000*. The design team believes that this vibration is associated with the drive shaft connecting the DC motor to the front-wheels of the car. The vibration manifests itself transiently when the DC motor is turned on/off and was found to be sufficiently large to be uncomfortable to the driver and passengers. The general conclusion from our meeting was that the magnitude of the vibration was unacceptable and needs to be reduced by at least 30%.

From discussions with team members of Mesa's *Gulo* design group, it appears that the "flexibility" of the drive-axle, which was assumed to be rigid in their design work, is the cause of the undesirable vibrations. Fortunately, we have available in-house, from previous work, an appropriate physical model of the motor and drive-axle system that has been instrumented and is accessible for your use in the Systems Lab. Your group is assigned the task of first developing a motor/drive-axle model to better understand the system and to subsequently remedy the vibration problem.

Specifically, you need to

- (1) validate that the vibration is indeed due to shaft flexibility,
- (2) characterize the system and develop a mathematical model, and
- (3) perform simulations using the validated model to identify possible design changes to solve the vibration problem.

Mesa would like to review several possible solutions to the vibration problem and select, based on a broad range of criteria beyond the actual drive shaft design, the best design for the final powertrain. Therefore, you are charged with identifying the four most effective solutions to achieve a 30% reduction in the vibrations.

This project has an extremely short timeline and must have your highest priority. You will be presenting the results of your experimental work and analysis in a poster session in the week of September 21. Because of the significance of this vibration problem, the poster session will be attended by Mesa engineers and executives with diverse technical backgrounds. Please ensure that your poster uses clear visual elements to provide a brief, broadly comprehensible introduction and conclusions alongside the central technical data. In addition, a short, three-page handout summarizing the key findings from your work is due to us in the same week during your poster presentation.

From discussions with team members of Mesa's *Gulo* design group, it appears that the "flexibility" of the drive-axle, which was assumed to be rigid in their design work, is the cause of the undesirable vibrations. Fortunately, we have available in-house, from previous work, an appropriate physical model of the motor and drive-axle system that has been instrumented and is accessible for your use in the Systems Lab. Your group is assigned the task of first developing a motor/drive-axle model to better understand the system and to subsequently remedy the vibration problem.

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ME 495- LABORATORY II

Lab 2 -- Frequency Response Method

Flexible Torsional System

Summary of Task Assignments:

Compare the predicted and measured frequency response

Determine/refine B_{tr} and B_{tl} or, perhaps better,

Determine B_m , B_s , B_d

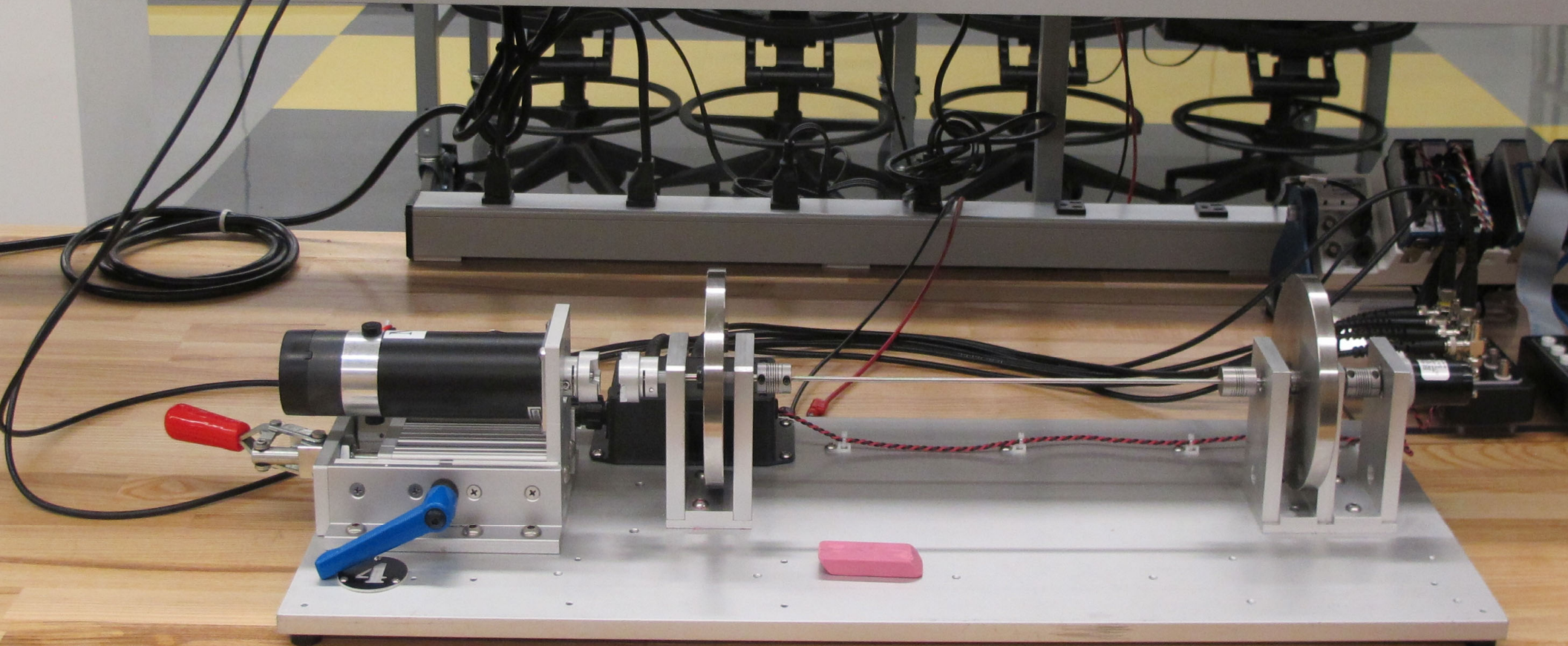
Model improvements: Determine improved model parameters – beyond the damping parameters – to obtain an acceptable (perhaps excellent) agreement between experiments and model

Perform a sensitivity analysis

Model design – Find 4 parameters for reducing the resonance

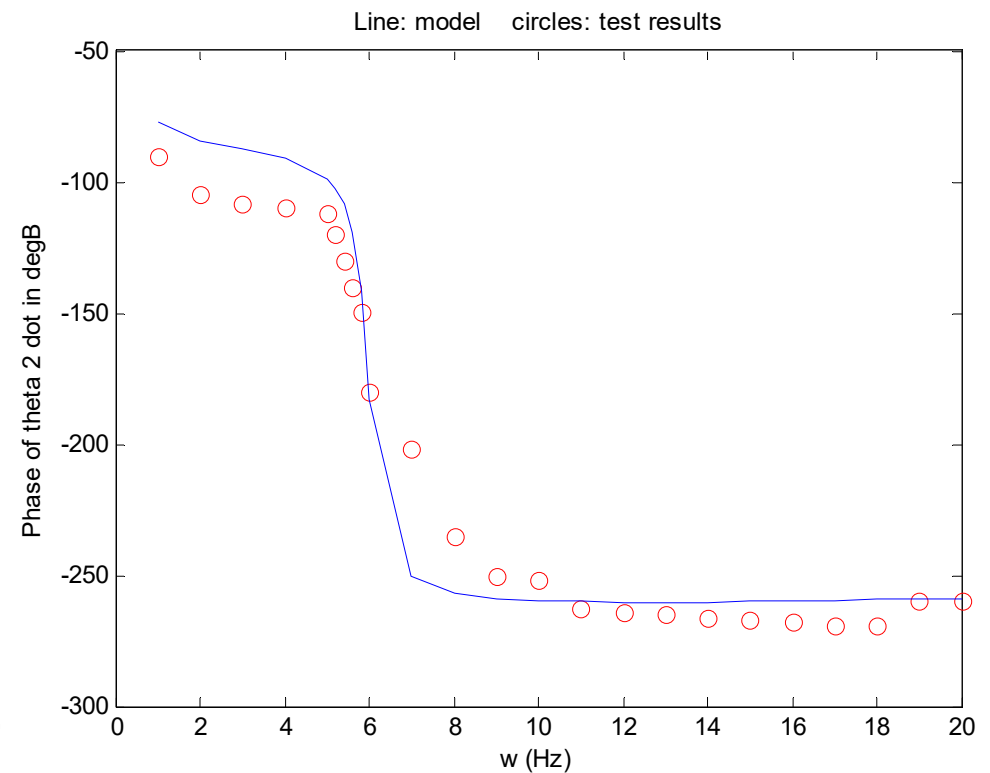
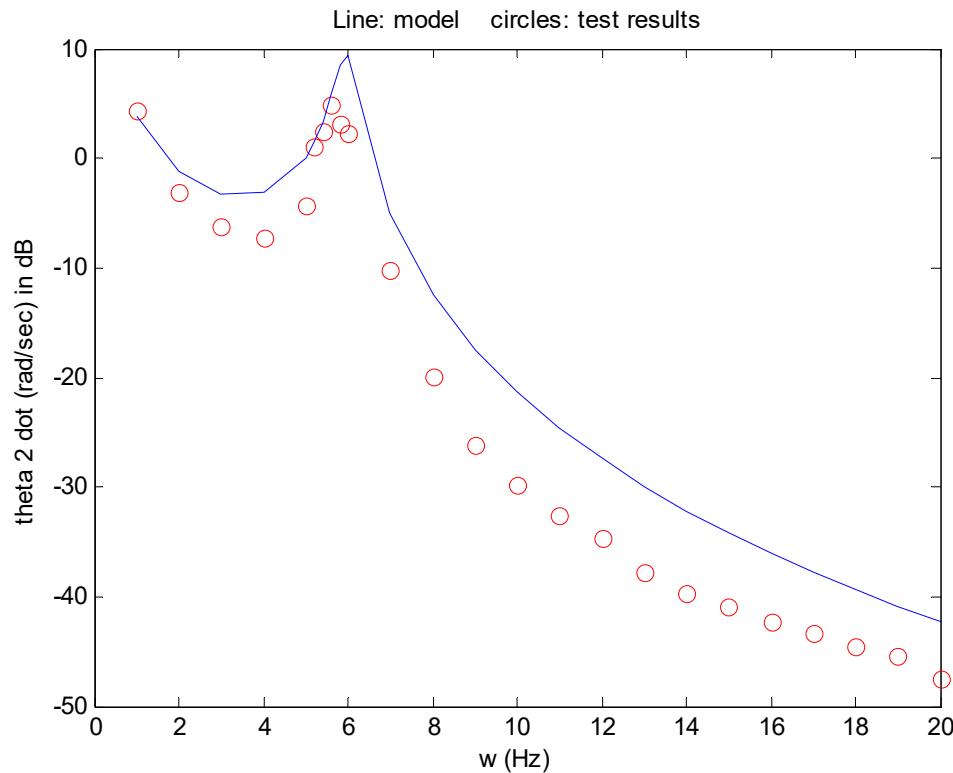
Model design – Perform in depth analysis for 4 parameters

Identify design changes to achieve goal (30% reduction in resonance)



Week 1 Results

- What do your results look like?
- Any concerns or issues?
- Did you obtain a good model?
- What is your plan of attack for the second week?



Code Plot

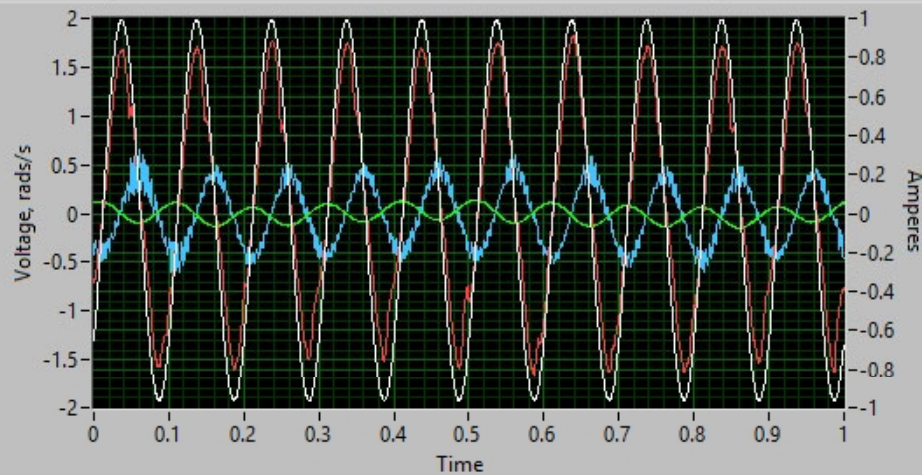
Frequency Response

Snapshot

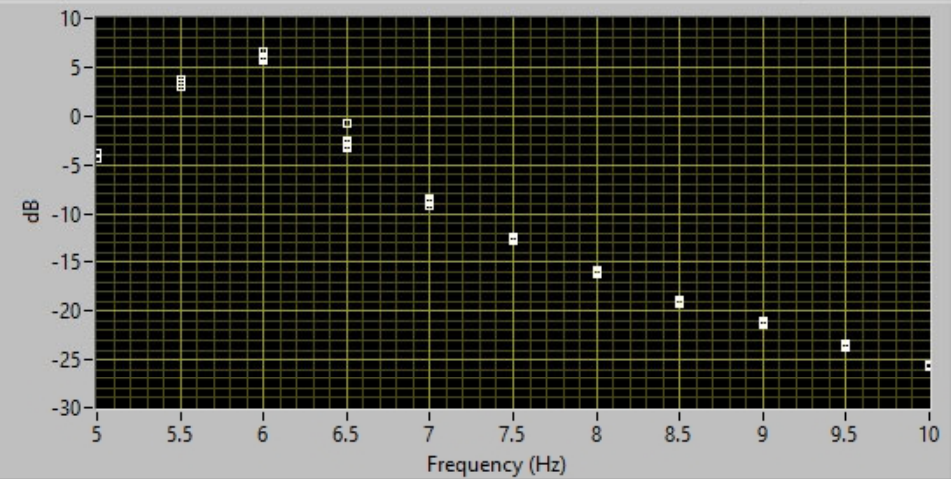
Configure

- ☒ Voltage
- ☒ Current
- ☒ Distal Tach
- ☒ Motor Tach

Waveform Graph



Bode Magnitude

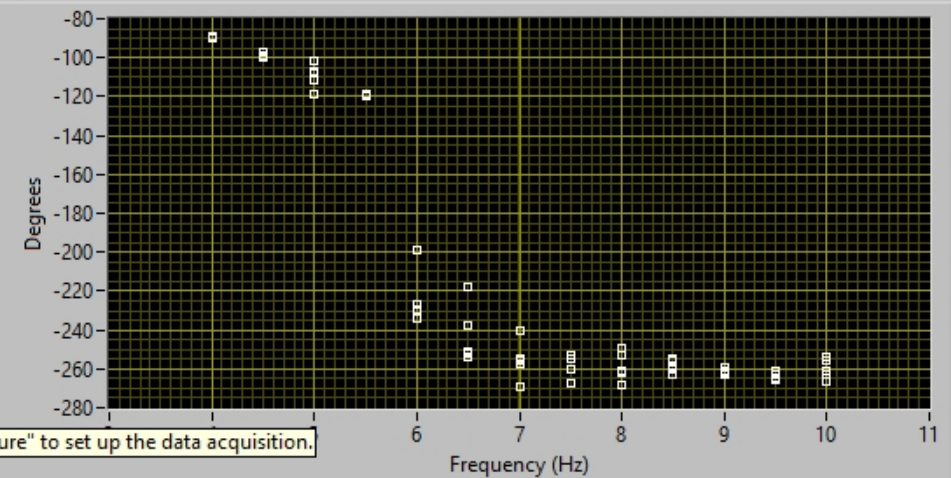


Current (A)

- ☒ Plot 0
- ☒ Plot 1



Bode Phase



Select "Take Data" to run a test and display the results. Select "Configure" to set up the data acquisition.

Damping Experiment

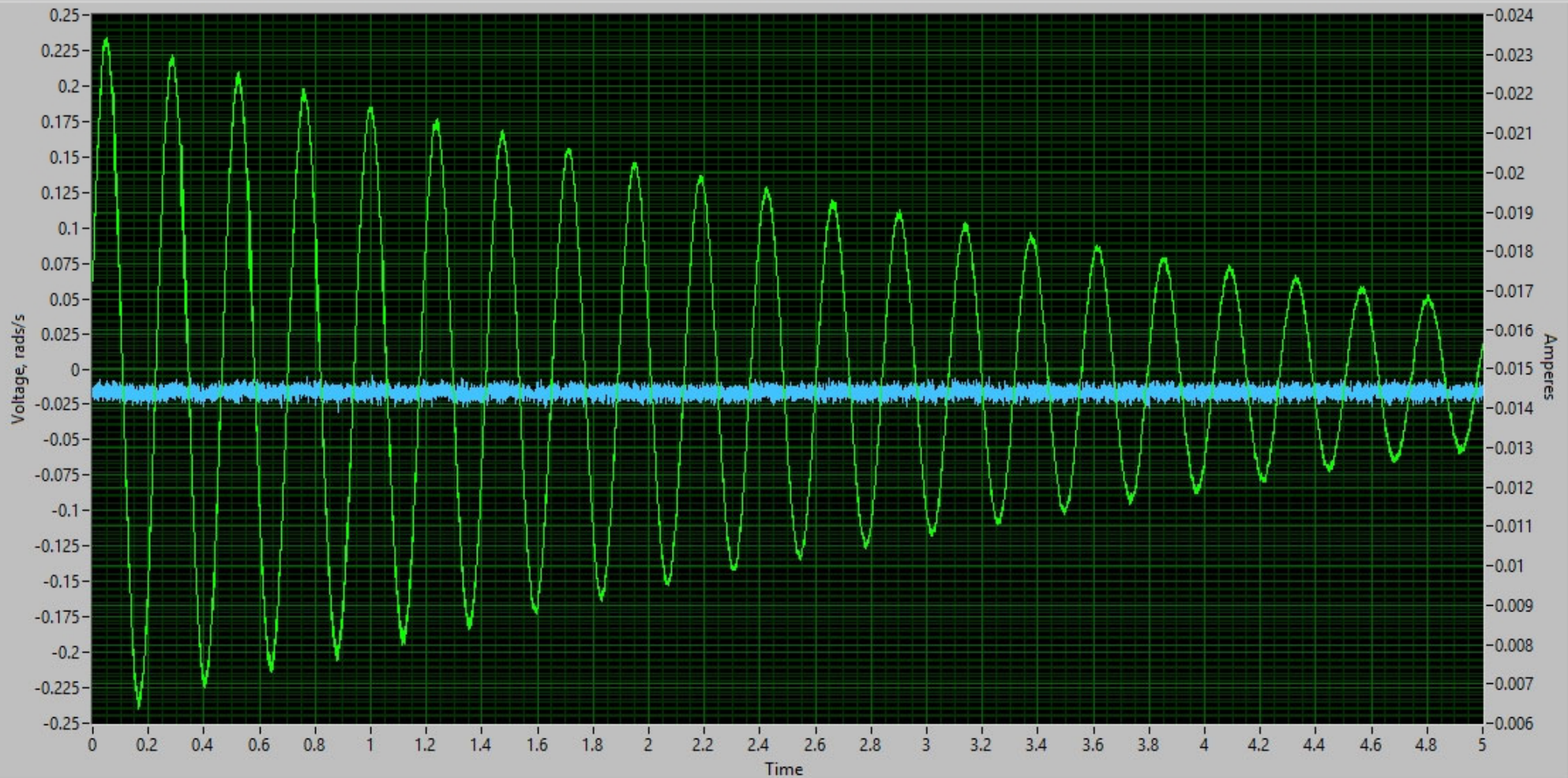
Frequency Response

Snapshot

Configure

- ☐ Voltage
- ☐ Current
- ☒ Tach Voltage
- ☒ Motor Tach

Waveform Graph 2



Week 2: Experimental and Analysis Tasks

- Repeat week 1 measurements as needed
- Vary frequency from high to low, noise method
- Determine impact of input amplitude
- Refine and improve model parameters
- Validate model
- Perform sensitivity analysis
- Identify the 4 parameters you propose to change in the design to reduce the resonance problem.
- Carry out an in-depth analysis. Are certain parameters linked?
- Summarize all quantitative results for your report

Model Usage (Improved Design)

MATLAB/SIMULINK example programs are available on the Course web page:

Example1.m Compare the predicted and measured frequency response

Example2.m Sensitivity analysis and cost calculations

Simscape See student guide. This is a toolbox in the SIMULINK editor for nonlinear simulations

Example1.m

- Key of raw parameters

```
% Construct the model using system parameters
K_m          = 0.06;          % N-m/A or V/(rad/sec)
R_m          = 0.9;           % Ohms
L_m          = 0.002;         % Henry
J_m          = 3.8e-5;        % Kg-m^2
D_flywheel   = 0.137;         % m
L_flywheel   = 0.0127;        % m
rho_flywheel = 7755;          % kg/m^3
D_shaft       = 3.18e-3;       % m
L_shaft       = 0.305;         % m
G_shaft       = 7.31e10;       % N/m^2
B_shaft       = 0.003;         % Type in your value here
B_bearing     = 0.003;         % Type in your value here
B_motor       = 0.00;         % Type in your value here
```

Example1.m (Cont'd)

Calculate computed parameters

```
% Computed parameters
```

```
J_f = rho_flywheel*pi*D_flywheel^4*L_flywheel/32;  
K   = pi*G_shaft*D_shaft^4/(32*L_shaft);  
J_c = J_m + J_f;
```

Calculate frequency response

```
num = [K_m*B_shaft, K_m*K];  
den = [R_m*J_c*J_f, R_m*J_c*B_tr+J_f*(K_m^2+B_tl*R_m),  
       R_m*(K*(J_c+J_f)+B_bearing*B_shaft)+  
       (K_m^2+R_m*B_motor)*B_tr, K*(K_m^2+B_tr*R_m)];  
[mag_model, phase_model] = bode(num,den,freq_test);
```

Transfer Function

$$H(s) = \frac{\dot{\theta}_2(s)}{V_{in}(s)}$$
$$= \frac{K_m(B_s s + K)}{R_m J_c J_f s^3 + [R_m J_c B_{tr} + J_f (K_m^2 + B_{tl} R_m)] s^2 + \{R_m [K(J_c + J_f) + B_b B_s] + (K_m^2 + R_m B_m) B_{tr}\} s + K [K_m^2 + (B_b + B_m) R_m]}$$

where

$$B_{tr} = B_s + B_b$$

$$B_{tl} = B_s + B_m$$

$$J_f = \frac{\rho_f \pi D_f^4 L_f}{32}$$

$$K = \frac{G_s J_s}{L_s} = \frac{\pi G_s D_s^4}{32 L_s}$$

Example1.m (Cont'd)

Calculate measured frequency response

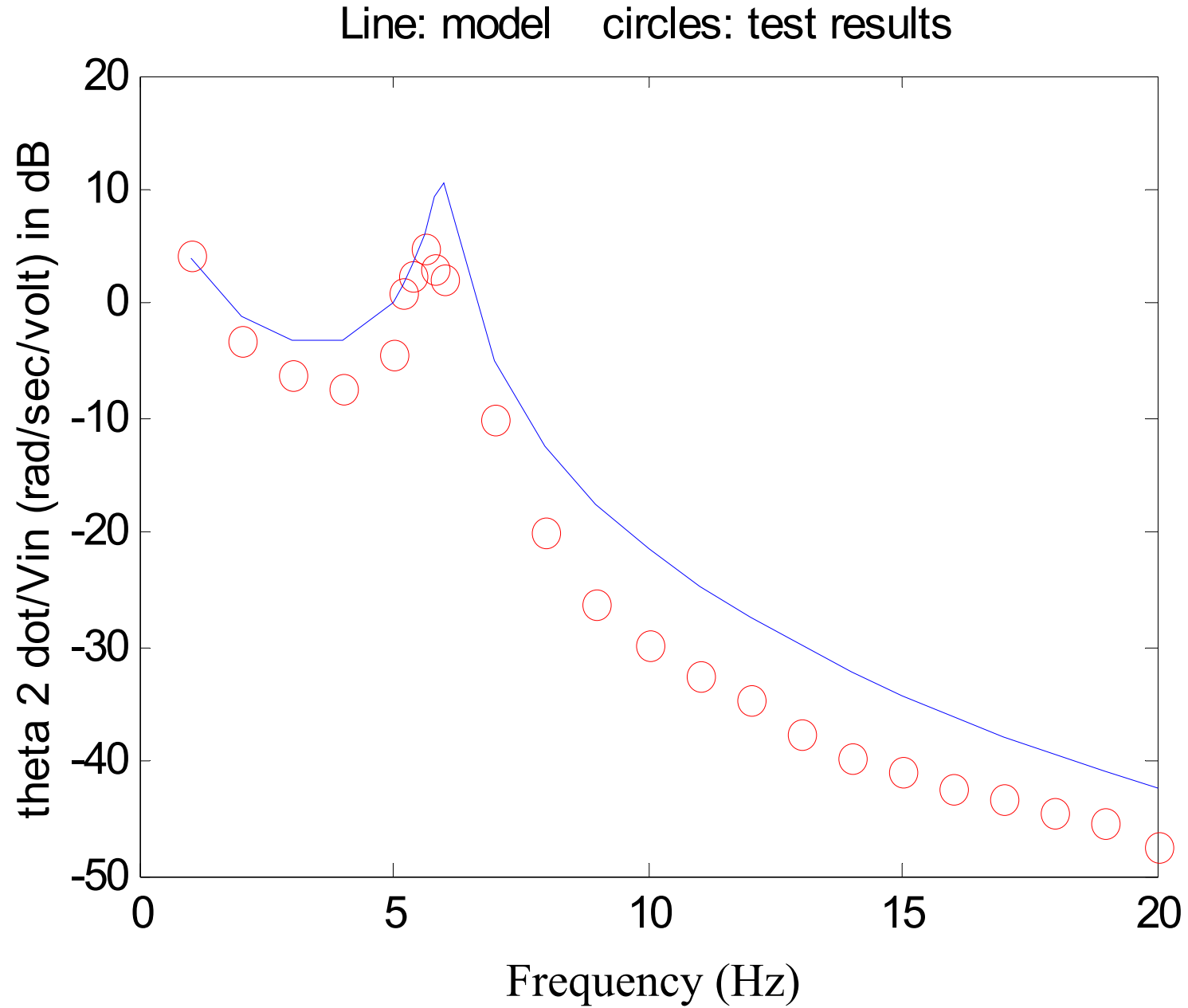
```
freq_test=2*pi*[1 2 3 4 5 5.2 5.4 5.6 5.8 6 ...  
7 8 9 10 11 12 13 14 15 16 17 18 19 20];  
mag_test = [ 1.18 0.496 0.344 0.308 0.432 0.800 0.940 1.240 ...  
1.020 0.920 0.220 0.072 0.035 0.023 0.0168 0.0132 ...  
0.0092 0.0074 0.0064 0.0054 0.0048 0.0042 0.0038  
0.003];  
  
mag_test = mag_test* 5.083; % Convert to rad/sec  
mag_test = mag_test/3.60; % Input voltage is  
3.60V p-p
```

Note: This is simply an example to get started. You must verify that this is correct for your work. Do not blindly trust!

Example1.m (Cont'd)

Compare model predicted and test results

```
plot(freq_test/(2*pi),20*log10(mag_test), 'ro',  
      freq_test/(2*pi), 20*log10(mag_model), 'b-')  
xlabel('w (Hz)')  
ylabel('theta 2 dot/Vin (rad/sec/volt) in dB')  
title('Line: model      circles: test results')  
figure(2)  
plot(freq_test/(2*pi),phase_test, 'ro',  
      freq_test/(2*pi), phase_model, 'b-')  
xlabel('w (Hz)')  
ylabel('Phase of theta 2 dot Transfer function in deg')  
title('Line: model      circles: test results')
```

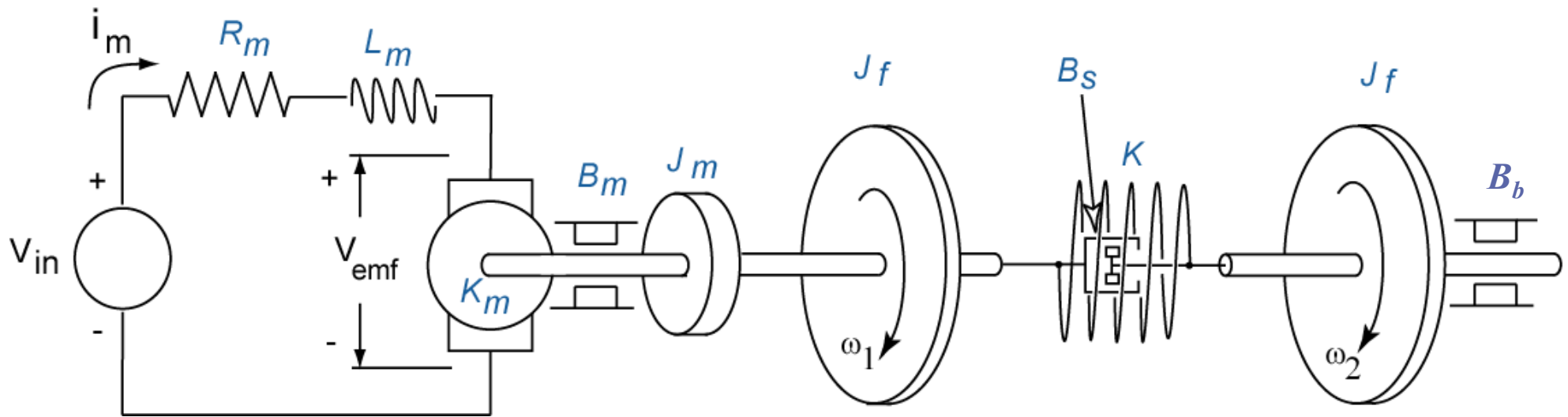


Model Adequacy

- We need an appropriate model that is quantitatively similar to the “plant” we are trying to simulate.
- Which is the most important region?
- What are the likely suspects if there appears to be a mismatch?
- You may leverage your model to explore this.

$$H(s) = \frac{\dot{\theta}_2(s)}{V_{in}(s)}$$
$$= \frac{K_m(B_s s + K)}{R_m J_c J_f s^3 + [R_m J_c B_{tr} + J_f(K_m^2 + B_{il} R_m)] s^2 + \{R_m[K(J_c + J_f) + B_b B_s] + (K_m^2 + R_m B_m) B_{tr}\} s + K[K_m^2 + (B_b + B_m) R_m]}$$

Model Adequacy



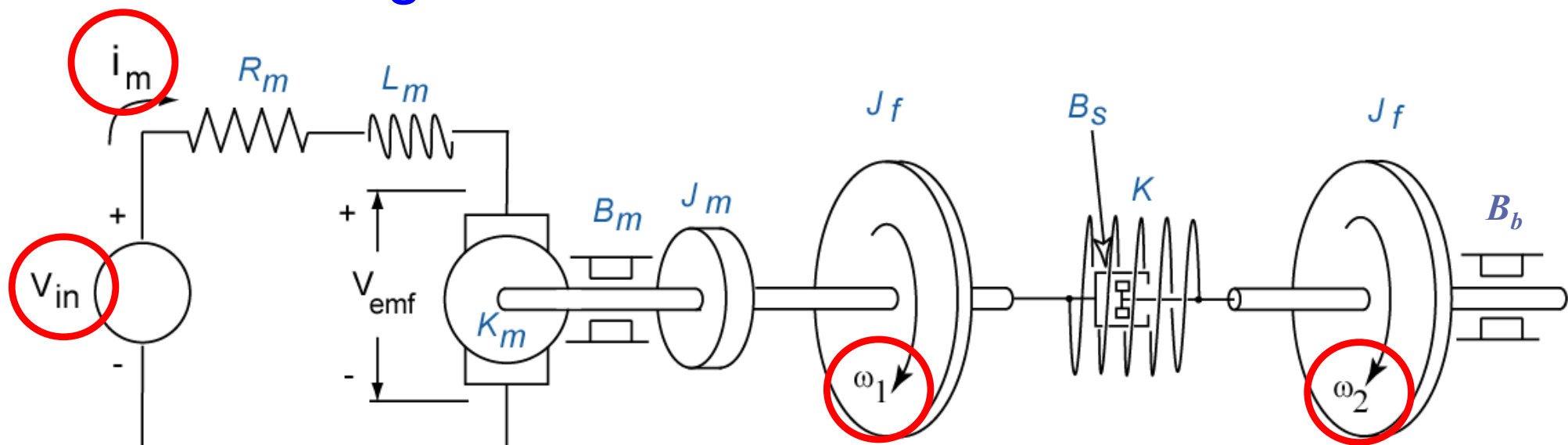
Given the discrepancy, which model parameters are likely responsible?

Obtain and Verify All Model Parameters

DC Motor		
Motor constant	$K_m = 0.097$ $K_m = 0.06$	MagMotor [N·m/Amp] or [V/(rad/sec)] Aerotech [N·m/Amp] or [V/(rad/sec)]
Winding resistance	$R_m = 1.6$ $R_m = 0.9$	MagMotor [ohms] Aerotech Motor
Winding inductance	$L_m = 0.002$	[Henry]
Rotor inertia	$J_m = 4.23 \times 10^{-5}$	MagMotor [kg·m ²]
2 Flywheels		
Diameter	$D_f = 0.137$	[m]
Length (width)	$L_f = 0.0127$	[m]
Density	$\rho_f = 7755$	[kg/m ³]
Shaft		
Diameter	$D_s = 3.18 \times 10^{-3}$	[m]
Length (width)	$L_s = 0.305$	[m]
Shear modulus	$G_s = 7.31 \times 10^{10}$	[N/m ²]
Tachometer #1		
Scale factor	7.0 (MagMotor) 3.0 (Aerotech)	[V/KRPM]
Tachometer #2		
Scale factor	20.8	[V/KRPM]

How Do We Obtain Improved Motor Parameters?

What signals are measured in the lab?



$$V_{in} = R_m i_m + V_{emf}$$

$$V_{in} = R_m i_m + K_m \dot{\theta}_1$$

$$V_{emf} = K_m \dot{\theta}_1$$

$$\tau_1 = K_m i_m$$

$$\tau_1 - \tau_2 - B_m \dot{\theta}_1 = (J_m + J_f) \ddot{\theta}_1$$

How Can We Determine Motor Parameters?

1. Estimate R_m

$$V_{in} = R_m i_m + K_m \dot{\theta}_1$$

At steady state, $\dot{\theta}_1 = \dot{\theta}_2$

And at high frequencies, motor is practically stationary ($\dot{\theta}_1 = 0$)

Thus, estimate R_m at high frequencies.

How Do We Compute Motor Parameters?

2. Evaluate K_m and R_m

$$V_{in} = R_m i_m + K_m \dot{\theta}_1$$

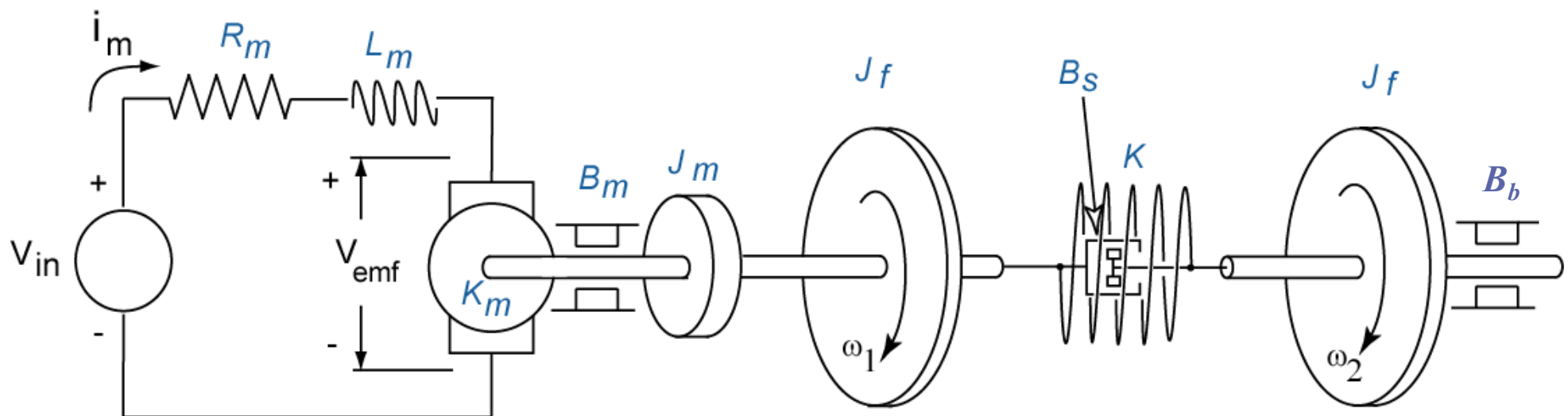
$$(V_{in} - R_m i_m) = K_m \dot{\theta}_1$$

Step 1. Operate motor at a range of (appropriate) frequencies, record parameters and plot the results.

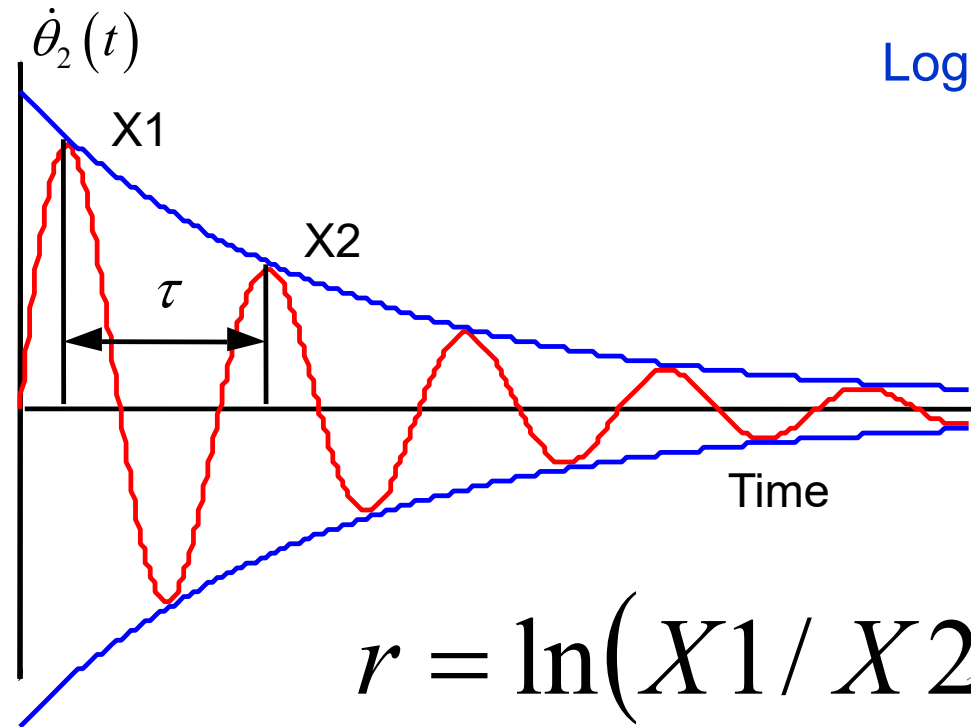
Step 2. Does this curve reveal interesting data about K_m and R_m ?

Estimate Damping Coefficients (started last week)

- Design a test procedure to obtain the damping coefficient B_{tl}
- In addition, you may want to design a test procedure to determine all three coefficients. If you choose to skip this step, you can assume $B_m = 0$ and attribute half of B_{tr} to B_s and the other half to B_b . You may choose other ways to divide up the damping, but please always justify your assumptions and measurement procedures.



Free Response of 2nd Order Systems



Log Decrement Method

$$\zeta = \frac{r}{\omega_n \tau}$$

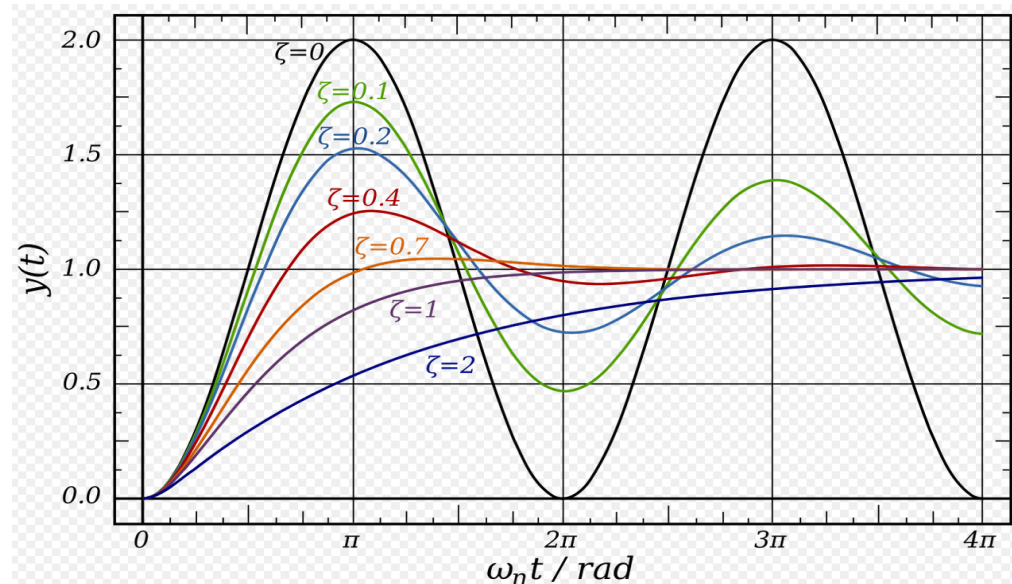
If there is little damping

$$\omega_n \tau \sim 2\pi$$

Determining a Damping Coefficient

$$\zeta = \frac{c \text{ (damping coefficient)}}{c_c \text{ (critical damping)}}$$

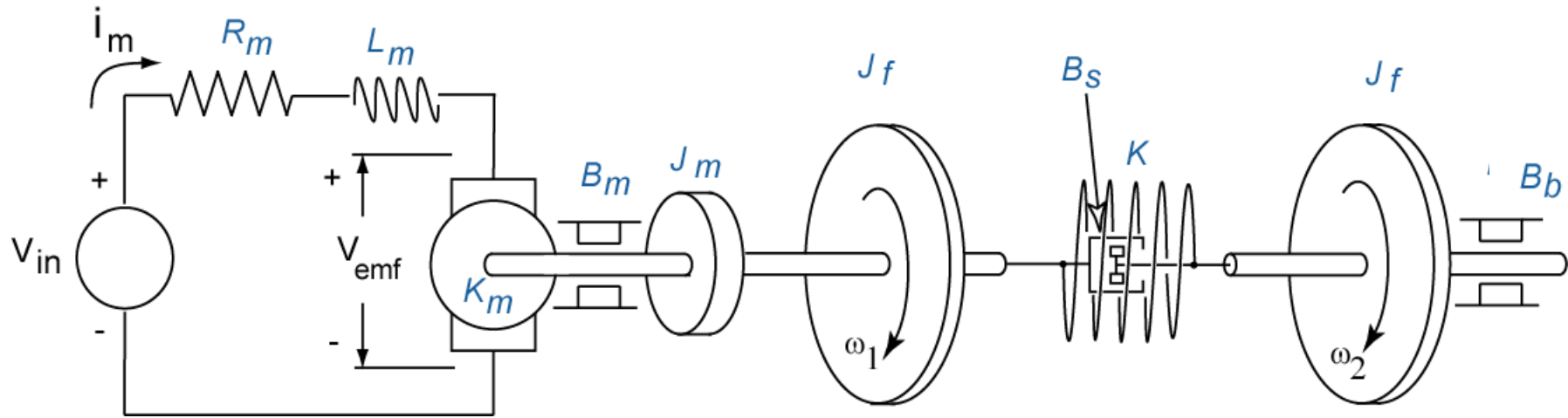
$$c_c = 2 \cdot \sqrt{k \cdot J}$$



Units: $c = \left[\frac{\text{kg m}^2}{\text{s}} \right]$

$\zeta = 1$ is critically damped

Determining (viscous) damping for ($B_m + B_b$)



Strategy: Measure the angular velocity as a function of time after the motor has been turned off

$$J \frac{d\omega}{dt} = \text{Torque} = -B_T \omega \quad \text{where} \quad B_T = B_m + B_b \quad \text{and} \quad J \text{ is the total moment of inertia.}$$

The solution to this equation is of the form: $\omega(t) = \omega(t=0)e^{-(B_T/J)t}$

This implies that the angular velocity will decay exponentially if the damping is viscous.

Is the damping viscous? Check Experimentally!

Strategy: If the damping is predominantly coulombic, the governing equation would be of the following form:

$$J \frac{d\omega}{dt} = \text{Torque} = -B_T \quad \text{where } B_T = B_m + B_b \quad \text{and } J \text{ is the total moment of inertia.}$$

The solution to this equation is of the form:

$$\omega(t) = \omega(t=0) - \left[\frac{B_T}{J} \right] t$$

This implies that the angular velocity would decay linearly.

Suggestions:

1. Understand the difference.
2. Decide how you want to represent dissipation in your model. Keep the tradeoff between a potential increase in model accuracy and complexity in mind.
3. Start with a simple model, increase the complexity only if it is needed for the work at hand.
4. Be prepared to justify your choice.

Sensitivity Analysis

Find the most effective measures in achieving the design goal. First, define a “cost function”, J , which is to be optimized (usually minimized). Sensitivity is then a measure of the change in cost due to parameter variations.

The sensitivity:

$$S_p^J \equiv \frac{\Delta J}{\Delta p} = \frac{\text{change in } J}{\text{change in } p}$$

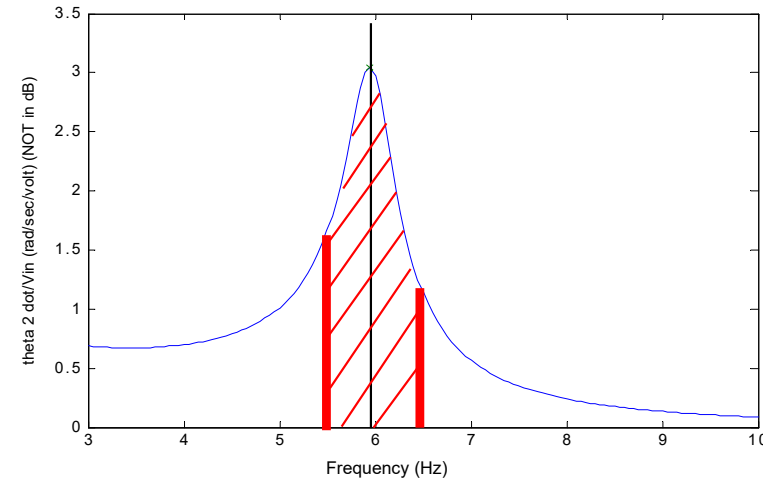
Sensitivity Analysis

Normalized sensitivity

$$S_p^J \equiv \frac{\Delta J / J}{\Delta p / p} = \frac{\text{percentage change in } J}{\text{percentage change in } p}$$

Cost Function for This Lab

$$J = \int_{\omega_n - \pi}^{\omega_n + \pi} \text{mag}\left[\frac{\dot{\theta}_2}{V_{in}}(j\omega)\right] d\omega$$



Meaning:

Objective is to reduce the “transmissibility” around the resonant frequency, measured in a 1Hz wide band.

Tasks:

- Compute original cost function.
- Perturb a parameter and compute new cost function.
- Compute the sensitivity.

Compute Original Cost Function

```
num_o = [K_m*B_shaft, K_m*K];  
den_o = [R_m*J_c*J_f, R_m*J_c*B_tr+J_f*(K_m^2+B_tl*R_m),  
         R_m*(K*(J_c+J_f)+B_bearing*B_shaft)+  
         (K_m^2+R_m*B_motor)*B_tr, K*(K_m^2+B_tr*R_m)];  
  
freq_FR=2*pi*[3:0.05:10]; % Assume resonance is between  
                           % 3 and 10 Hz  
  
[mag_o1 phase_o1] = bode(num_o,den_o,freq_FR);  
[max_mag_o1, index1] = max(mag_o1);  
resonance_frq = freq_FR(index1);  
  
min_freq = resonance_frq - pi; % Minimum freq in rad/sec  
max_freq = resonance_frq + pi; % Maximum freq in rad/sec  
freq_COST = min_freq: (pi)/100 : max_freq;  
  
[mag_o2 phase_o2] = bode(num_o,den_o,freq_COST);  
cost_o = sum(mag_o2); % This is a 'sum' but is a good  
                     % approximate of integral
```

Perturb a Parameter and Compute New Cost Function

```
B_shaft = B_shaft*1.01;  
% Update all the computed parameters  
J_f = rho_flywheel*pi*D_flywheel^4*L_flywheel/32;  
.....
```

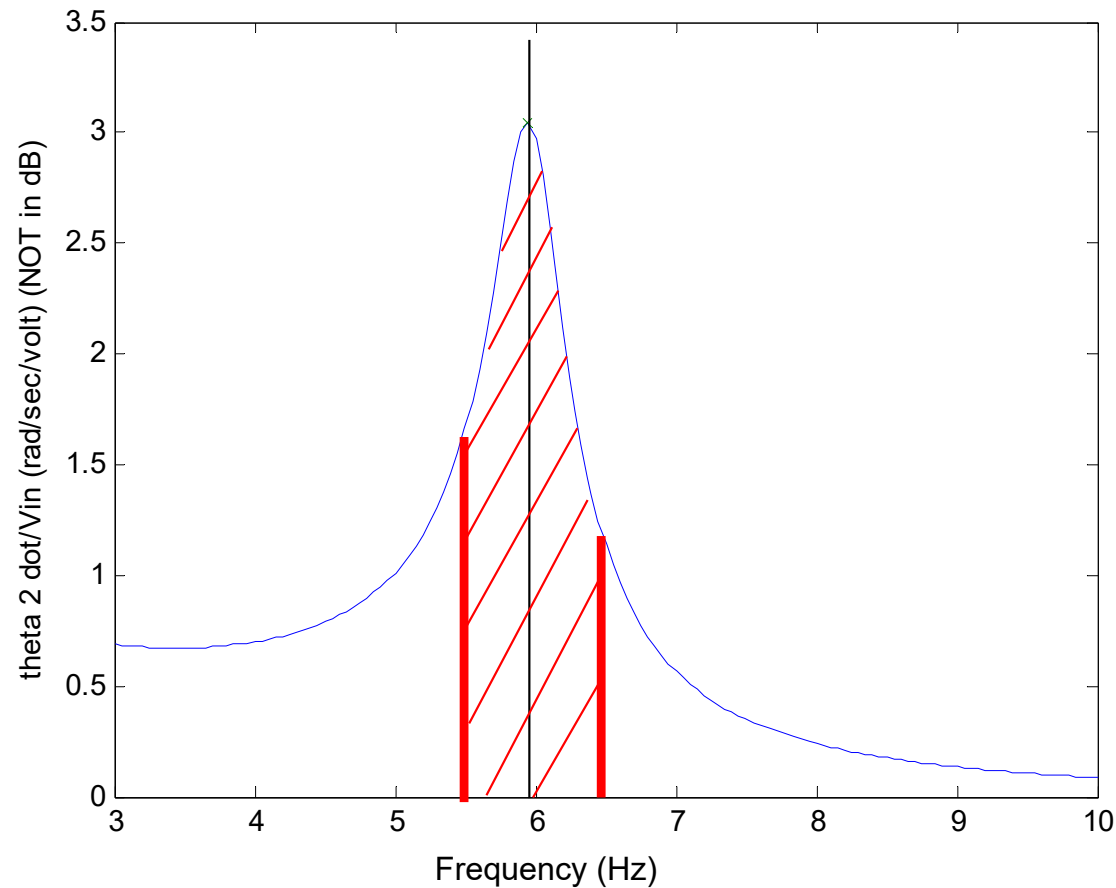
```
% Update Transfer function
```

```
% Compute new frequency response
```

```
% Calculate the new cost
```

```
% Undo the change
```

Repeat for both positive and negative perturbations
And find the normalized sensitivity by averaging.



>> example2

ans =

With 1% parameter perturbation, the cost function changes by 0.3266%

New Designs

- Find the **4** most effective parameters to achieve the design goal (to reduce resonance).
- Perform more in-depth analysis only for the 4 selected parameters (due to limited resources).
- **Identify design changes** that will achieve the goal (**30% reduction in resonance**).
- Summarize your proposed changes.
- Notice: Active control solutions are not considered in this specific project, but are certainly feasible.

ME 495- LABORATORY II

Flexible Torsional System

Summary of Task Assignments:

Measured frequency response and compare to predictions

Determine B_{tr} and B_{tl}

Model development: refine model parameters and demonstrate model adequacy.

Perform a sensitivity analysis

Model design – Find 4 parameters for reducing resonance (i.e. the transmissibility at resonance)

Model design – Perform in depth analysis for 4 parameters

Identify design changes to achieve goal (30% reduction in resonance)

ME 495- LABORATORY II

Consider for superior work and presentation, based on this week's lab work:

Determine B_m , B_s , B_b or included a better way and/or discussion to describe damping and friction

Consider a non-linear model id needed

Modeling and characterization of motor behavior

Vary input amplitudes

Vary frequency from high to low (noise analysis?)

Confirm given model parameters and/or refine calibrations for improving your model and experimental results

Planning Your Poster and Handout

Prepare a detailed outline!

- Ultimately, the outline must define the poster at the level of every section and paragraph.
- Every section must have a purpose, goal, results and/or conclusions.
- Every section needs an introductory heading or sentence, a Figure or main body and should be leading over to the next topic.
- Sections should be logically connected. This logical flow (introduction, questions & tasks, methods, results, discussion, conclusions) ultimately supports the points you want to make.

A good team member will not allow a group to begin with the poster and handout unless everybody agrees on the completed outline.

Planning Your Poster and Handout - Outline

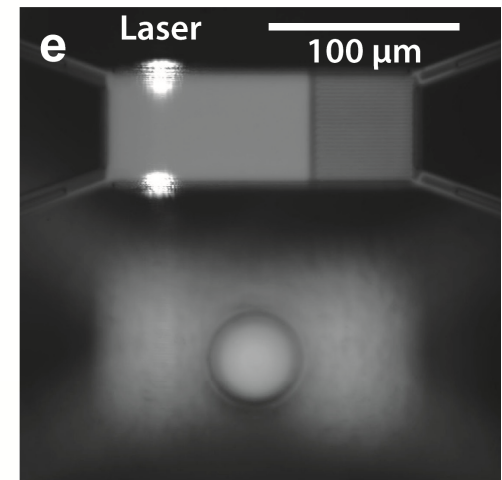
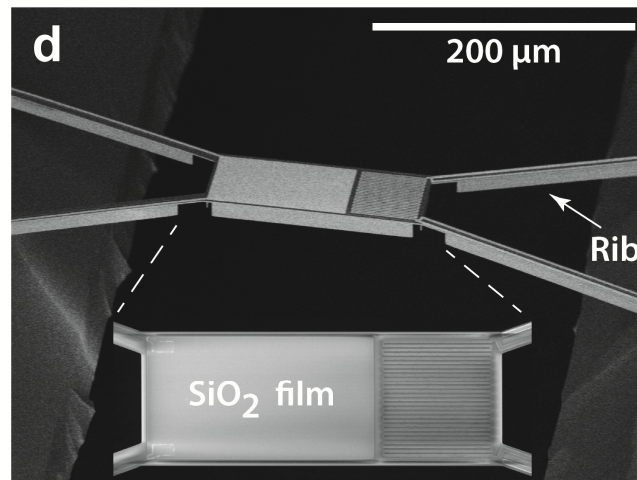
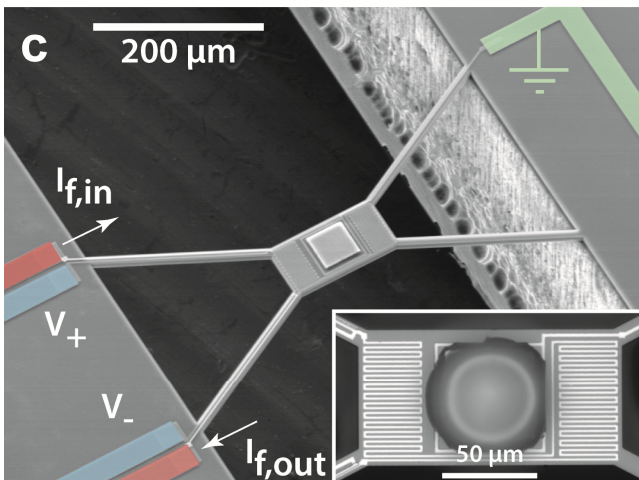
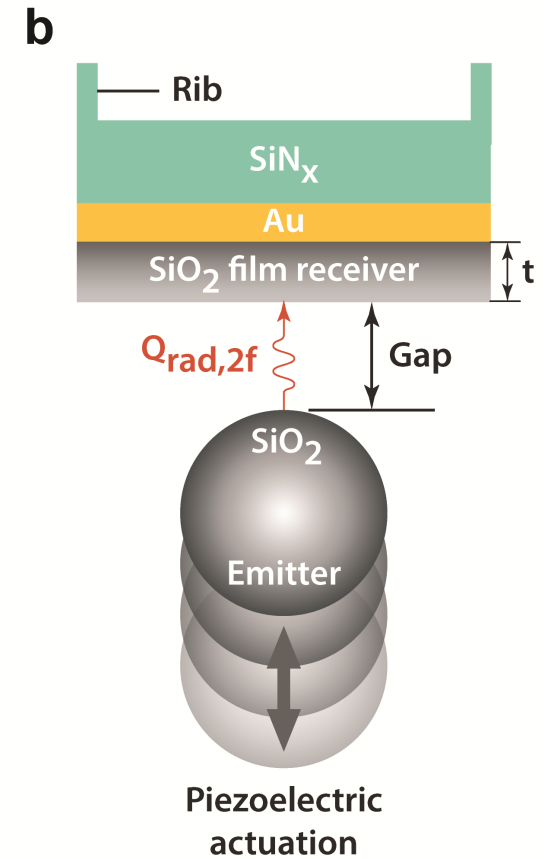
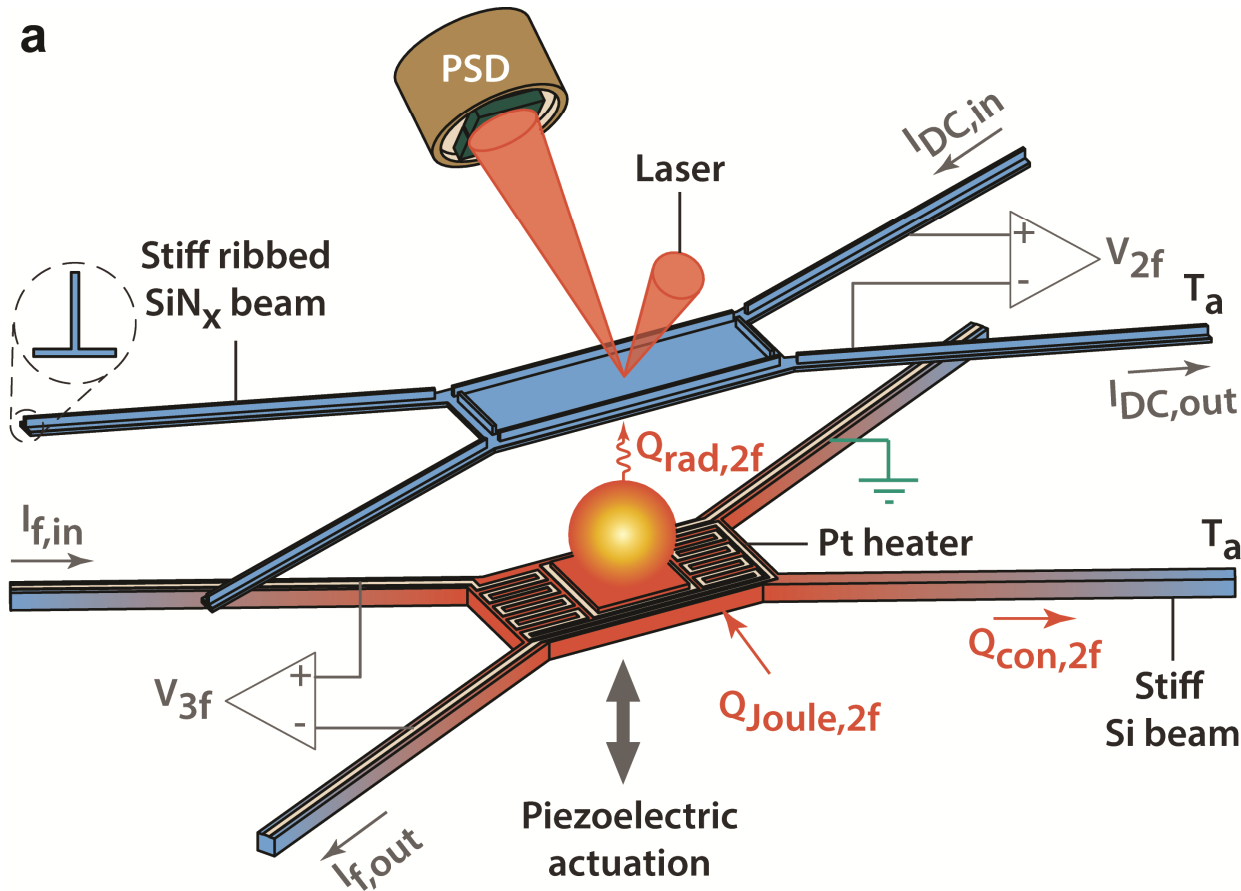
- Review the goals of your project. Establish a systematic order of all goals.
- Start with the Results section.
- Prepare draft plots of your results. Outline the sequence in which you present the results.
- Outline the Materials and Methods used.
- Outline the “Recommendation or Solutions” section. Carefully consider which question(s) you need to address and provide clear answers and recommendations. Progress towards the more complex.
- Draft the Final Conclusion(s) and Abstract (Goals) at the end.
- Outline the Introduction.

Begin work only if your outline is complete!

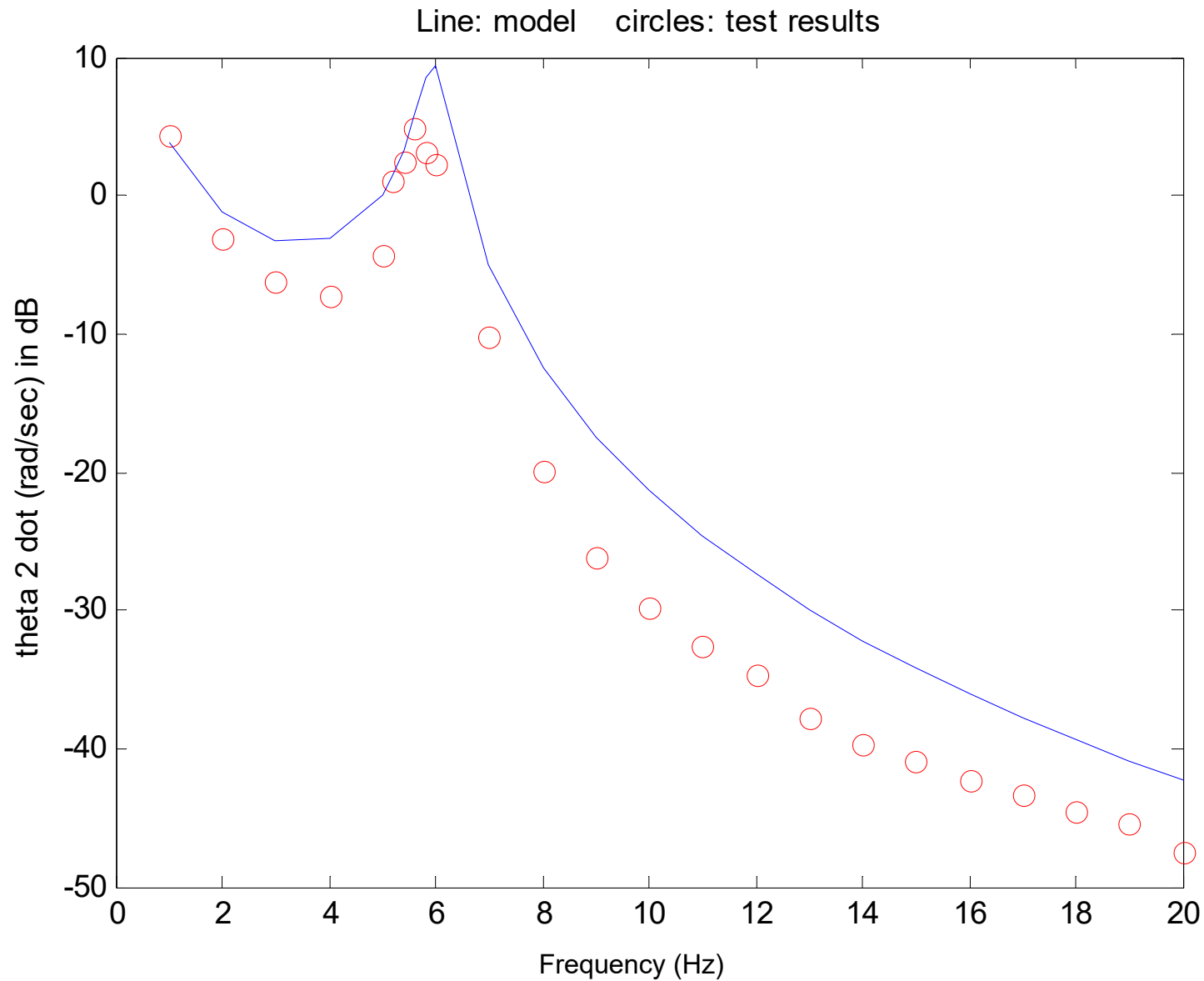
What is Expected from Your Poster/Handout

- Address the technical questions in **TASK LETTER**.
- **Be prepared to answers technical questions underpinning the results and conclusions shown!**
- Clear **FIGURES**, clearly visible and large, distinguishable symbols, lines, labels etc.
EXCEL typically yields poor figures, use Origin Pro and Illustrator (available to all)
- **Don't state opinions as facts**. Identify your opinions as such with phrases such as “*We believe...*”.
- **State your assumptions**, make clear to the audience (reader) what your assumptions are.
Do these assumptions limit your conclusion?
“Based on our assumption of we conclude that”
Relax the assumptions and describe how this effects your conclusions.
- **Keep the language simple**, and the purpose clear (say why you did it). No one is impressed with complicated syntax. Everyone is impressed with a clear presentation.
- **Don't try to draw conclusions where you have no clear data, knowledge or insight**.
This damages your credibility. When in doubt, be conservative with your conclusions.
Mark speculative as speculative.
- Suggesting **design changes** and making recommendations:
 - Make sure the proposed design changes are **possible and reasonable**.
 - Present your suggested changes as if you were an objective outsider (i.e. they can be defended objectively using your results, data and modeling).
- **Don't “puff up”** your report (all sections need to be defensible and add information, value).
- Practice your presentation often and thoroughly; **10 minutes** is just 10 minutes!
- Answer questions **concisely and to the point**.
- Most common issue: **WHAT ARE THE UNITS?**
- Audience: Ask questions **professionally**.

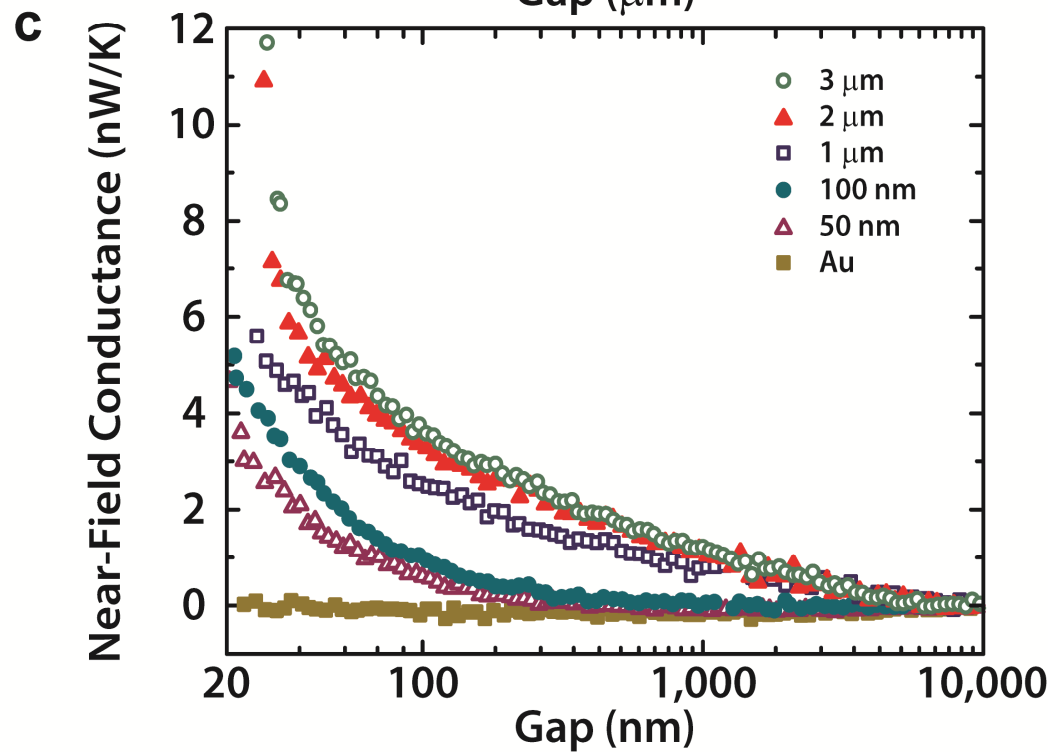
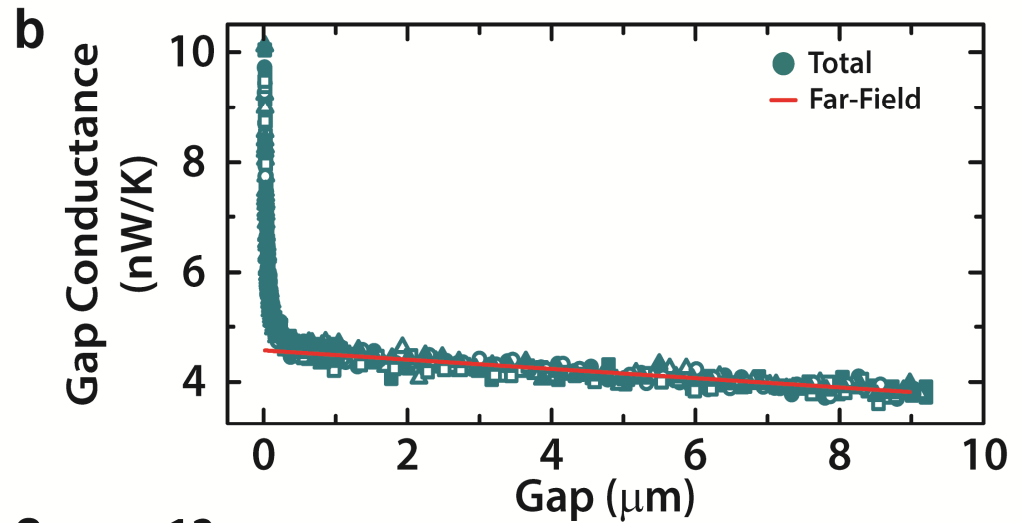
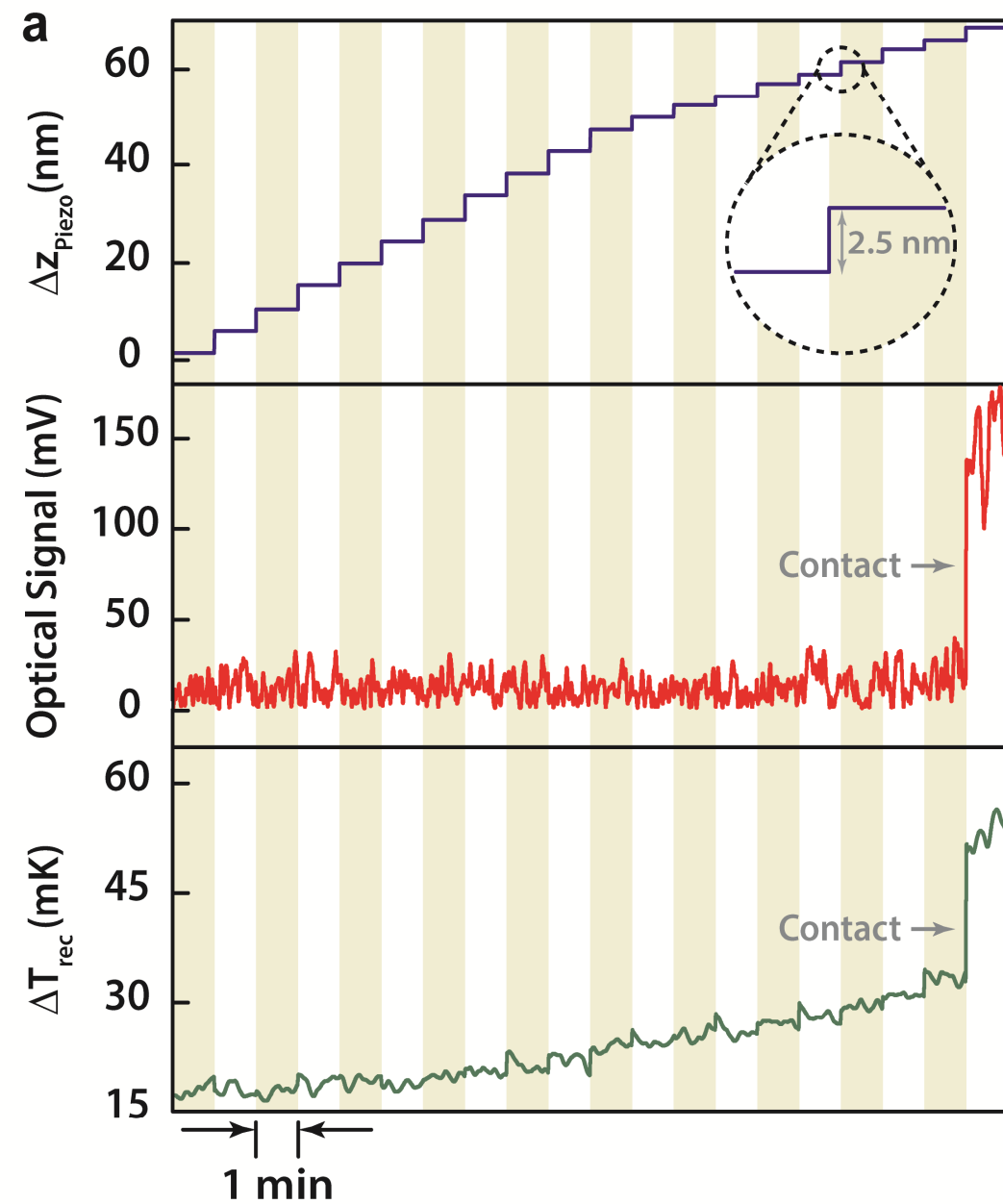
Sketches are Superior to Explain Complex Engineering Measurements



Is this a Quality Plot?



Develop Clear Data Figures



Poster Presentation

- Poster Presentations are in person in GGBL 2541.
- Bring 4 printed hardcopies of your poster (8.5x11 inch) and handout. Turn them in at the beginning of your presentation.
- Upload your poster (ppt or pdf file) and handout files prior to your poster session.
- In addition, email the poster and the handout to your GSI no less than 30 minutes prior to your presentation.
- All teams and team members will attend the entire ~ 2 hour-long presentation session and be both presenters and audience.
- All team members should present – we are practicing an exceptionally important skill. Exceptions can be requested under some circumstances, but this must be done at least 24 h in advance via email to Prof. Meyhofer.
- Please ensure that all team members contribute equally.
- Poster presentations should be **max. 10 minutes long**. After 11 minutes your presentation has to (and will) stop.
- Q&A after each presentation for about 10 - 15 minutes with audience questions (technical and communication).
- Presentation times (starting times organized by your respective lab times):

Tu:	11:30 am	3:20 pm	5:30 pm
W:	11:30 am	3:20 pm	5:30 pm
Th:		3:20 pm	5:30 pm
- Practice your presentation thoroughly and repeatedly. **10 minutes** is just 10 minutes!
- **No reading** of text from phone, note cards etc.
- Answer questions **concisely and to the point**. Consider in advance who will handle which questions.
- Ask questions **professionally**.
- During all presentations your GSI, a tech comm. instructor, and Profs Reddy and Meyhofer will be present.
- No lectures next week. Use the time for **preparation, practice and appointments for additional office hours**.
- Additional office hours will be held on Monday 1:30 – 3 pm.